

CSE4261: Neural Network and Deep Learning

Assignment-10

Name: Md Sajjad Hossain

ID: 2011176125

Session: 2019-20

July 18, 2025

1. Generating Images from Noise using Autoencoder

We used the decoder of a fully connected Autoencoder trained on CIFAR-10. The encoder maps input images to a 128-dimensional latent vector. To evaluate its generative capacity, we sampled five random vectors from a standard normal distribution ($\mu = 5$, $\sigma = 1$) and passed them through the decoder.

Observation: The generated images were blurry and lacked meaningful content. This outcome is expected because the latent space of a standard Autoencoder is not designed for random sampling — it is only optimized for reconstruction.

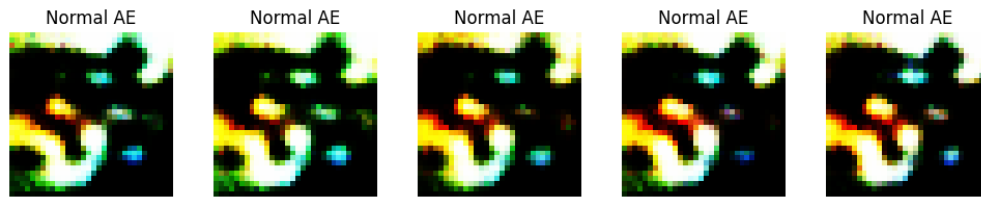


Figure 1: Generated images from 5 noise vectors using the Autoencoder (Mean=5, Std=1)

2. Generating Images using Denoising Autoencoder

We trained a similar Autoencoder with a denoising objective: input images were corrupted with Gaussian noise ($\mu = 5$, $\sigma = 1$), and the model was trained to recover clean images. After training, we sampled 5 random latent vectors from $\mathcal{N}(5, 1)$ and passed them through the decoder.

Observation: The generated outputs were slightly smoother compared to the standard Autoencoder, but still visually meaningless. Since the latent space was not regularized for generative purposes, the decoder could not map random vectors to realistic images.

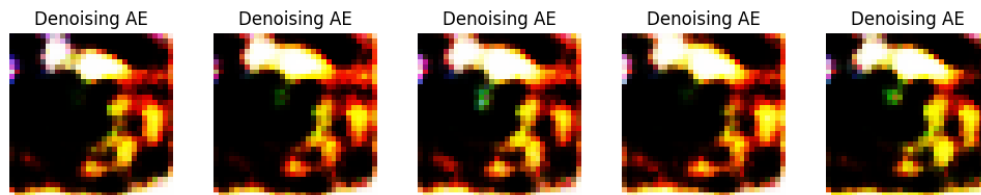


Figure 2: Generated images from 5 noise vectors using the Denoising Autoencoder (Mean=5, Std=1)

3. Training and Evaluation of Variational Autoencoder (VAE)

We implemented and trained a convolutional Variational Autoencoder (ConvVAE) on CIFAR-10. The encoder outputs a mean and log-variance vector of size 128, and sampling is done using the reparameterization trick. The model was trained for 20 epochs using a combination of reconstruction loss and KL divergence.

Observations:

- Loss values printed after each epoch showed gradual convergence of both reconstruction and KL terms.
- The reconstructed images appeared smoother and more semantically accurate than those produced by standard or denoising autoencoders.
- The decoder generated visually reasonable outputs, although the images were a little blurry.

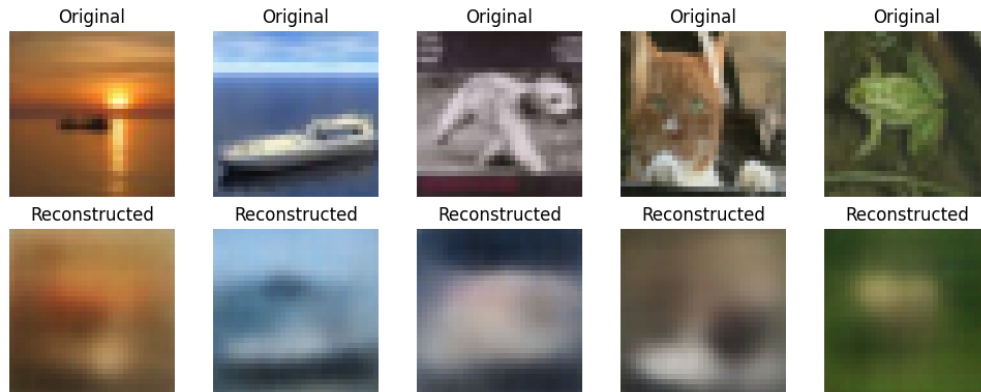


Figure 3: Reconstruction results using the trained Variational Autoencoder

4. Generating Images from Noise using Variational Autoencoder

To test generative capability, we sampled 5 random latent vectors from $\mathcal{N}(5,1)$ and passed them through the decoder of the trained VAE.

Observations:

- The generated images had distinct structures, with object-like features such as edges and color patches.
- Although the outputs were not perfect, they were more coherent compared to those from the standard and denoising autoencoders.
- This shows that the VAE decoder has learned a sufficiently smooth latent space.

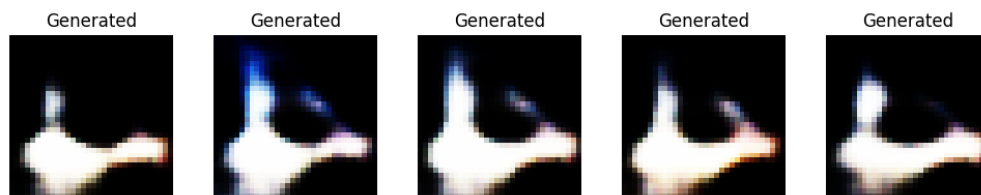


Figure 4: Generated images from 5 noise vectors using the trained VAE (Mean=5, Std=1)

Conclusion

Our experiment highlighted the limitations of traditional and denoising autoencoders in generative settings. Their latent spaces are not structured for sampling, resulting in meaningless outputs. In contrast, the Variational Autoencoder, through probabilistic modeling and KL regularization, learned a continuous and smooth latent space, enabling realistic image generation from random noise vectors.

Code Link:

Here's the github link of the implemented code: <https://github.com/mdsajjadhossain25/CSE4261>